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NONVOLATILE MEMORY DEVICE, METHOD OF OPERATING THE SAME, AND MEMORY SYSTEM INCLUDING THE SAME

Abstract

A nonvolatile memory device may include processing circuitry configured to receive a read command, a start read address, and a data chunk indicator, the read command including a page read command and a data read command, and the start read address including a start row address and a start column address, a memory cell array configured to output first page data based on the start row address, the first page data including a plurality of first data chunks, and the processing circuitry further configured to output one or more of the plurality of first data chunks based on the data chunk indicator.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This U.S. non-provisional application claims the benefit of priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0028052 filed on Feb. 27, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

[0002] Various example embodiments of the inventive concepts described herein relate to a semiconductor device, and more particularly, relate to a nonvolatile memory device, an operating method of the nonvolatile memory device, and/or a memory system including the nonvolatile memory device, etc.

[0003] Electronic devices which include a plurality of memory systems including nonvolatile memory devices are widely used. Such memory systems support stability and durability, fast access speed of information, and consume less power.

[0004] The nonvolatile memory device may include a memory cell array, a page buffer circuit, and a data input/output circuit. In the process of reading data stored in the memory cell array, the memory cell array may output page data to the page buffer circuit, and the page buffer circuit may output a part of or all of the page data to the data input/output circuit. When the page buffer circuit outputs a part of the page data, more commands may be desired and/or required depending on the structure of the page buffer circuit in which the part of the page data is stored. In electronic devices including a plurality of nonvolatile memory devices, increases in the number of commands may create a bottleneck in an interface circuit, which may act as an overhead of the interface circuit and may degrade input/output performance of the nonvolatile memory device, decrease the access speed of the information, increase the power consumption of the electronic device, etc.

SUMMARY

[0005] One or more example embodiments of the inventive concepts provide a nonvolatile memory device which improves input/output performance.

[0006] One or more example embodiments of the inventive concepts provide an operating method of a nonvolatile memory device.

[0007] One or more example embodiments of the inventive concepts provide a memory system which includes a nonvolatile memory device.

[0008] According to at least one example embodiment, a nonvolatile memory device includes processing circuitry configured to receive a read command, a start read address, and a data chunk indicator, the read command including a page read command and a data read command, and the start read address including a start row address and a start column address, a memory cell array configured to output first page data based on the start row address, the first page data including a plurality of first data chunks, and the processing circuitry further configured to output one or more of the plurality of first data chunks based on the data chunk indicator.

[0009] According to at least one example embodiment, an operating method of a nonvolatile memory device may include receiving a read command, a start read address, and a data chunk indicator, the read command including a page read command and a data read command, and the start read address including a start row address and a start column address, outputting first page data based on the start row address, the first page data including a plurality of first data chunks, and outputting one or more of the plurality of first data chunks based on the data chunk indicator.

[0010] According to at least one example embodiment, a memory system includes a memory controller configured to output a read command, a start read address, and a data chunk indicator,

the read command including a page read command and a data read command, and the start read address including a start row address and a start column address, and a nonvolatile memory device configured to operate based on the read command, the start read address, and the data chunk indicator, the nonvolatile memory device including, a memory cell array configured to output first page data based on the start row address, the first page data including a plurality of first data chunks, and a page buffer circuit configured to continuously output two or more of the plurality of first data chunks based on the data chunk indicator.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0011] The above and other objects and/or features of the example embodiments of the inventive concepts will become apparent by describing in detail some example embodiments thereof with reference to the accompanying drawings.

[0012] FIG. 1 is a block diagram illustrating a memory system including a nonvolatile memory device according to at least one example embodiment of the inventive concepts.

[0013] FIG. 2 is a diagram explaining planes, blocks, and pages included in a memory cell array of FIG. 1 according to at least one example embodiment.

[0014] FIG. 3 is a diagram explaining a start read address and a data chunk indicator of FIG. 1 according to at least one example embodiment.

[0015] FIG. 4 is a diagram explaining one or more of a plurality of first data chunks output from a page buffer circuit of FIG. 1 based on a data chunk indicator according to at least one example embodiment.

[0016] FIG. 5 is a flowchart illustrating an operating method of a nonvolatile memory device according to at least one example embodiment of the inventive concepts.

[0017] FIG. 6 is a block diagram illustrating data and other control signals between a nonvolatile memory device of FIG. 1 and a memory controller communicating therewith according to at least one example embodiment.

[0018] FIG. 7 is a block diagram illustrating at least one example embodiment of a nonvolatile memory device of FIG. 1.

[0019] FIG. 8 is a timing diagram explaining at least one example embodiment of an operation of the nonvolatile memory device of FIG. 1, FIG. 6, or FIG. 7.

[0020] FIG. 9 is a flowchart explaining an operation in which a control circuit of FIG. 7 receives a read command, a start read address, and a data chunk indicator according to at least one example embodiment.

[0021] FIG. 10 is a flowchart illustrating at least one example embodiment of an operation in which a control circuit of FIG. 7 receives a read command and a start read address.

[0022] FIG. 11 is a flowchart illustrating at least one example embodiment of an operation in which a control circuit of FIG. 7 receives a read command and a data chunk indicator.

[0023] FIG. 12A and FIG. 12B are diagrams illustrating at least one example embodiment of a data chunk indicator of FIG. 1.

[0024] FIG. 13 is a flowchart illustrating at least one example embodiment of an operation in which a control circuit of FIG. 7 outputs one or more of a plurality of first data chunks.

[0025] FIG. 14 is a timing diagram explaining at least one example embodiment of an operation of the nonvolatile memory device of FIG. 7.

[0026] FIG. 15 is a block diagram illustrating at least one example embodiment of a memory controller communicating with a nonvolatile memory device of FIG. 1.

[0027] FIG. 16 is a flowchart illustrating an operating method of a nonvolatile memory device according to at least one example embodiment of the inventive concepts.

[0028] FIG. 17 is a block diagram illustrating a data center including another nonvolatile memory device according to at least one example embodiment of the inventive concepts.

DETAILED DESCRIPTION

[0029] Hereinafter, various example embodiments of the inventive concepts will be described clearly and in detail so that a person of ordinary skill in the technical field of the inventive concepts may practice the example embodiments of the inventive concepts.

[0030] FIG. 1 is a block diagram illustrating a memory system including a nonvolatile memory device according to at least one example embodiment of the inventive concepts.

[0031] Referring to FIG. 1, a memory system **10** may include a memory controller **100** and a nonvolatile memory device **300**. The nonvolatile memory device **300** may include a control circuit **310**, a memory cell array **330**, and/or a page buffer circuit **350**, etc., but is not limited thereto and for example, the memory system **10** may include a greater or lesser number of constituent components. According to some example embodiments, one or more of the memory controller **100**, the control circuit **310**, and/or the page buffer circuit **350**, etc., may be implemented as processing circuitry. The processing circuitry may include hardware or hardware circuit including logic circuits; a hardware/software combination such as a processor executing software and/or firmware; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc., but is not limited thereto.

[0032] The memory controller **100** may generally control operations of the nonvolatile memory device **300**, but is not limited thereto. For example, the memory controller **100** may store data in the memory cell array **330** of the nonvolatile memory device **300**, may read data stored in the memory cell array **330**, and/or erase data stored in the memory cell array **330**, etc. The storing of data, the reading of data, and/or the erasing of data may be initiated by an external host device and/or by the memory controller **100** itself.

[0033] The control circuit **310** may generally control operations of the memory cell array **330** and/or the page buffer circuit **350**, etc. For example, the memory cell array **330** may include a plurality of memory cells which store data, and the page buffer circuit **350** may include a plurality of latches which temporarily store data provided from the memory cell array **330** and/or the memory controller **100**, etc. The control circuit **310** may control the memory cells and the latches to perform the storing, reading, and/or erasing of data, etc.

[0034] When the memory controller **100** attempts to read specific data (hereinafter referred to as “target data”) stored in specific memory cells (hereinafter referred to as “target memory cells”) of the memory cell array **330**, a read command RDCMD, a start read address StrRDADR, and/or a data chunk indicator DCNKIND are generated, and the read command RDCMD, the start read address StrRDADR, and/or the data chunk indicator DCNKIND may be issued (and/or output, etc.) to the nonvolatile memory device **300**.

[0035] The nonvolatile memory device **300** may receive the read command RDCMD, the start read address StrRDADR, and/or the data chunk indicator DCNKIND from the memory controller **100**, and may provide the target data to the memory controller **100** based on the read command RDCMD, the start read address StrRDADR, and/or the data chunk indicator DCNKIND.

[0036] In at least one example embodiment, the read command RDCMD may include a page read command PRDCMD and a data read command DRDCMD, and the start read address StrRDADR may include a start row address StrRA and a start column address StrCA, but the example embodiments are not limited thereto. The page read command PRDCMD may be a command for outputting page data stored in one page of the memory cell array **330** from the memory cell array **330** to the page buffer circuit **350**, and the data read command DRDCMD may be a command for outputting some or all of the page data from the page buffer circuit **350** to a data input/output

circuit (not illustrated).

[0037] In at least one example embodiment, some or all of the output of the page data based on the data read command DRDCMD may be performed in units of data chunk (e.g., data chunk units, etc.). For example, one data chunk unit may be a set of data defined by dividing the page data evenly or unevenly (e.g., the size of a page of data may be evenly divisible by the size of a data chunk unit, or the size of a page of data may be unevenly divisible by the size of a data chunk unit), and the data chunk indicator DCNKIND may individually indicate each of the data chunks which are some or all of the page data and may be output from the page buffer circuit **350**. In other words, the size of one data chunk may be smaller than or equal to the size of a page of data. Further, the size of one data chunk may be variable and/or configurable, and may be defined and/or set using the data chunk indicator and/or a chunk size indicator to be discussed in greater detail in connection with FIG. **16**. Moreover, the size of one data chunk may be dynamically defined and/or dynamically set by a host device (e.g., an external device, etc.), the control circuit **310**, and/or the memory controller **100**, etc., using the data chunk indicator and/or a chunk size indicator, but the example embodiments are not limited thereto.

[0038] In at least one example embodiment, the start row address StrRA and the start column address StrCA may indicate the page data output from the memory cell array **330** to the page buffer circuit **350**, and a location indicated by the start row address StrRA and the start column address StrCA may be a reference for the data chunk indicator DCNKIND to individually indicate and/or specify each of the data chunks. The read command RDCMD, the start read address StrRDADR, and the data chunk indicator DCNKIND will be described in greater detail with reference to FIG. **3**, FIG. **8**, FIG. **12A**, and FIG. **12B**.

[0039] In at least one example embodiment, the control circuit **310** receives the read command RDCMD, the start read address StrRDADR, and/or the data chunk indicator DCNKIND from the memory controller **100**, and may control the memory cell array **330** and the page buffer circuit **350**, respectively, based on the read command RDCMD, the start read address StrRDADR, and/or the data chunk indicator DCNKIND, etc. The memory cell array **330** may output first page data PGDAT1 including a plurality of first data chunks DCNK11, DCNK12, DCNK13, and the like to the page buffer circuit **350** based on the start row address StrRA. The page buffer circuit **350** may generate and/or transmit output data ODAT1 including one or more of the plurality of first data chunks DCNK11, DCNK12, DCNK13, and the like to the data input/output circuit based on the data chunk indicator DCNKIND, but the example embodiments are not limited thereto.

[0040] In at least one example embodiment, the first page data PGDAT1 may correspond to the start row address StrRA and may be data stored in memory cells arranged in one row of the memory cell array **330**, but is not limited thereto.

[0041] In at least one example embodiment, although the first page data PGDAT1 includes the plurality of first data chunks, e.g., DCNK11, DCNK12, DCNK13, and the like, the memory cell array **330** only outputs the page data stored in the memory cells arranged in one row of the memory cell array **330**, and therefore, the memory cell array may not output the first page data PGDAT1 in units of data chunk, but the example embodiments are not limited thereto. For example, in some example embodiments of the inventive concepts, the memory cell array **330** may perform normal operations of a conventional memory cell array, but is not limited thereto.

[0042] According to at least one example embodiment, data chunks (e.g., DCNKa, . . .) output from the page buffer circuit **350** may correspond to a plurality of read address ranges. The plurality of read address ranges may represent addresses of the target memory cells of the memory cell array **330** and may be determined by the host device and/or the memory controller **100** to start reading the target data, but is not limited thereto.

[0043] In at least one example embodiment, the plurality of read address ranges may be continuous (e.g., sequential) or discontinuous (e.g., non-sequential) in an address space of the memory cell array **330**. When the plurality of read address ranges are continuous (e.g., sequential, etc.), the data

chunks output from the page buffer circuit **350** may be stored in continuous latches of the page buffer circuit **350** (e.g., may be continuously stored, etc.), and when the plurality of read address ranges are discontinuous, the data chunks output from the page buffer circuit **350** may be stored in discontinuous latches of the page buffer circuit **350** (e.g., may be stored discontinuously, etc.).

[0044] In at least one example embodiment, the page buffer circuit **350** may output and/or continuously output two or more of the plurality of first data chunks **DCNK11**, **DCNK12**, **DCNK13**, and the like, which are indicated by the data chunk indicator **DCNKIND**. For example, the page buffer circuit **350** may output and/or continuously output data chunks regardless of whether the plurality of read address ranges are continuous or discontinuous. The plurality of read address ranges will be described below with reference to **FIG. 4**.

[0045] In at least one example embodiment, the page read command **PRDCMD**, the start row address **StrRA**, and/or the start column address **StrCA** may be defined in a toggle double data rate (**DDR**) standard associated with a flash memory interface for high-performance applications, but the example embodiments are not limited thereto. Unlike the page read command **PRDCMD**, the start row address **StrRA**, and/or the start column address **StrCA**, the data read command **DRDCMD** may not be defined in the toggle **DDR** standard and may be defined individually and/or specially defined in one or more example embodiments of the inventive concepts. For example, the page read command **PRDCMD** may include “00h” and “30h”, but is not limited thereto. For example, the data read command **DRDCMD** may not include “05h” and “E0h”, but is not limited thereto.

[0046] In at least one example embodiment, although the page read command **PRDCMD** is defined in the toggle **DDR** standard, the number and/or timing of issuing the page read command **PRDCMD** in some example embodiments of the inventive concepts may be variously modified differently from those which are defined in the toggle **DDR** standard, but the example embodiments are not limited thereto.

[0047] According to one or more example embodiments of the inventive concepts, with the above configuration, the nonvolatile memory device may continuously output a plurality of target data targeted for a data read operation in units of data chunks based on the data chunk indicator, and the size of the target data may be some or all of the size of a page of data. Thus, additional signals (e.g., additional commands and/or additional addresses) need not be generated, intervened, and/or applied, etc., between one data chunk and another data chunk output from the nonvolatile memory device, etc. The nonvolatile memory device may reduce, remove, and/or prevent the generation and/or intervention of additional signals, and thus the input/output performance of the nonvolatile memory device and/or a memory system may be improved, may be more efficient, etc.

[0048] **FIG. 2** is a diagram explaining planes, blocks, and pages included in a memory cell array of **FIG. 1** according to at least one example embodiment.

[0049] Referring to **FIG. 1** and **FIG. 2**, a memory cell array **330a** may correspond to the memory cell array **330**, but the example embodiments are not limited thereto.

[0050] The memory cell array **330a** may include a plurality of planes, e.g., **PL1** and **PL2**, etc. The plane **PL1** may include a plurality of blocks, e.g., **BLK11** and **BLK12**, etc., and the plane **PL2** may include a plurality of blocks, e.g., **BLK21** and **BLK22**, etc., but are not limited thereto. The block **BLK11** may include a plurality of pages, e.g., **PG11a**, **PG11b**, **PG11c**, and **PG11d**, etc., and the block **BLK12** may include a plurality of pages, e.g., **PG12a**, **PG12b**, **PG12c**, and **PG12d**, etc., but are not limited thereto. The block **BLK21** may include a plurality of pages, e.g., **PG21a**, **PG21b**, **PG21c**, and **PG21d**, etc., and the block **BLK22** may include a plurality of pages, e.g., **PG22a**, **PG22b**, **PG22c**, and **PG22d**, etc.

[0051] In at least one example embodiment, one of the plurality of pages **PG11a** to **PG11d**, **PG12a** to **PG12d**, **PG21a** to **PG21d**, and **PG22a** to **PG22d** may include target memory cells of the memory cell array **330a** for and/or associated with a read operation of the nonvolatile memory device **300**, and may store and/or contain the first page data **PGDAT1**. Accordingly, the memory cell array **330a** may output the first page data **PGDAT1** based on the start row address **StrRA**, and the page buffer

circuit **350** may output one or more (e.g., some or all) of the plurality of first data chunks DCNK11, DCNK12, DCNK13, and the like, based on the data chunk indicator DCNKIND. However, the example embodiments of the inventive concepts are not limited thereto. In another example embodiment, data chunks may be defined with respect to one block or one plane rather than one page (or greater than one page). When data chunks are defined for one block, the data chunks may be referred to as “a plurality of first block data chunks”, and when data chunks are defined for one plane, the data chunks may be referred to as “a plurality of first plane data chunks”, etc. The plurality of first block data chunks or the plurality of first plane data chunks may be respectively processed in the same as or similar manner to the plurality of first data chunks DCNK11, DCNK12, DCNK13, and the like. Accordingly, the page buffer circuit **350** may continuously output one or more of the plurality of first block data chunks or one or more of the plurality of plane data chunks, etc. Processing of the plurality of first block data chunks or the plurality of first plane data chunks according to some example embodiments of the inventive concepts will be described in greater detail later with reference to FIG. 14.

[0052] In FIG. 2, the memory cell array **330a** includes two planes PL1 and PL2, and each plane (e.g., PL1) includes two blocks (e.g., BLK11 and BLK12), and each block (e.g., BLK11) includes four pages (e.g., PG11a, PG11b, PG11c, and PG11d), but the number of planes, blocks, and pages included in the memory cell array **330a** are illustrated as examples and are not limited thereto.

[0053] FIG. 3 is a diagram explaining a start read address and a data chunk indicator of FIG. 1 according to at least one example embodiment.

[0054] Referring to FIG. 1 and FIG. 3, the memory cell array **330** may output first page data PGDAT1x based on the start read address StrRDADR.

[0055] In at least one example embodiment, the start read address StrRDADR may include the start row address StrRA and the start column address StrCA, and the start row address StrRA and the start column address StrCA may indicate the first page data PGDAT1x stored in a page PGx. The first page data PGDAT1x may include a plurality of first data chunks DCNK11x, DCNK12x, DCNK13x, DCNK14x, DCNK15x, DCNK16x, DCNK17x, and DCNK18x, but is not limited thereto. For example, the start read address StrRDADR may indicate the start of the page PGx and/or start of the data chunk DCNK11x (e.g., **401**), but the example embodiments of the inventive concepts are not limited thereto.

[0056] Based on the data chunk indicator DCNKIND, the page buffer circuit **350** may output a plurality of data chunks, e.g., DCNK12x, DCNK14x, and DCNK16x, etc., among the plurality of first data chunks DCNK11x to DCNK18x.

[0057] In at least one example embodiment, the data chunk indicator DCNKIND may indicate the plurality of data chunks DCNK12x, DCNK14x, and DCNK16x, etc., from all of the plurality of first data chunks DCNK11x to DCNK18x (e.g., **403**), and the page buffer circuit **350** may generate and/or transmit output data ODAT1x including the data chunks DCNK12x, DCNK14x, and DCNK16x, etc. As described above with reference to FIG. 1, the data chunks DCNK12x, DCNK14x, and DCNK16x may be continuously output, but is not limited thereto.

[0058] FIG. 4 is a diagram explaining one or more of a plurality of first data chunks output from the page buffer circuit of FIG. 1 based on a data chunk indicator according to at least one example embodiment.

[0059] In FIG. 4, a plurality of read address ranges RDADRRNGs and pages PG(x-1), PGx, PG(x+1) (where x is an integer of 2 or more) are illustrated. The plurality of read address ranges RDADRRNGs may indicate address ranges (e.g., ADDRNG1, ADDRNG2, and ADDRNG3, etc.) for reading target data from target memory cells of a memory cell array, which are targets for reading data. The pages PG(x-1), PGx, PG(x+1) may be adjacent to each other in the memory cell array, and the page PGx may include the target memory cells for and/or to be used in storing the target data. For example, the plurality of read address ranges RDADRRNGs may include two or more discontinuous address ranges of the memory cell array, but is not limited thereto.

[0060] Referring to FIG. 4, an address range ADDRNG1 may correspond to a data chunk DCNK12x, an address range ADDRNG2 may correspond to a data chunk DCNK14x, and an address range ADDRNG3 may correspond to a data chunk DCNK16x, but the example embodiments are not limited thereto.

[0061] In FIG. 4, there may be a gap between the address ranges ADDRNG1 and ADDRNG2 and/or between the address ranges ADDRNG2 and ADDRNG3, which may indicate that the address ranges ADDRNG1 to ADDRNG3 are discontinuous in an address space of the memory cell array. In this case, the data chunks DCNK12x, DCNK14x, and DCNK16x may also be discontinuously stored in the first page data PGDAT1x (and/or may be discontinuously stored in latches of the page buffer circuit).

[0062] As described above with reference to FIG. 1 and FIG. 3, the relationship between the plurality of read address ranges RDADDRNGs and the data chunks DCNK12x, DCNK14x, and DCNK16x may be identified by the data chunk indicator DCNKIND regardless of whether the address ranges are continuous or discontinuous.

[0063] FIG. 5 is a flowchart illustrating an operating method of a nonvolatile memory device according to at least one example embodiment of the inventive concepts.

[0064] Referring to FIG. 1 and FIG. 5, the control circuit 310 may receive the read command RDCMD, the start read address StrRDADR, and/or the data chunk indicator DCNKIND (S100), but is not limited thereto. The read command RDCMD may include the page read command PRDCMD and the data read command DRDCMD, and the start read address StrRDADR may include the start row address StrRA and the start column address StrCA, but are not limited thereto.

[0065] The memory cell array 330 may output the first page data PGDAT1 including the plurality of first data chunks, e.g., DCNK11, DCNK12, DCNK13, and the like, based on the start row address StrRA (S300).

[0066] The page buffer circuit 350 may output one or more (e.g., DCNKa, . . .) of the plurality of first data chunks, e.g., DCNK11, DCNK12, DCNK13, and the like, based on the data chunk indicator DCNKIND (S500).

[0067] FIG. 6 is a block diagram explaining data and other control signals between the nonvolatile memory device of FIG. 1 and the memory controller communicating therewith according to at least one example embodiment.

[0068] Referring to FIG. 6, a memory system 500 may correspond to the memory system 10 of FIG. 1, but is not limited thereto, and may include at least one memory controller 510 and at least one nonvolatile memory device 530, etc., but the example embodiments are not limited thereto.

[0069] The nonvolatile memory device 530 may correspond to one of a plurality of nonvolatile memory devices communicating with the memory controller 510 based on one or more of a plurality of channels. The memory controller 510 may correspond to the memory controller 100 of FIG. 1, and the memory device 530 may correspond to the nonvolatile memory device 300 of FIG. 1, but the example embodiments are not limited thereto.

[0070] The memory device 530 may include a plurality of pins, e.g., first to eighth pins P11 to P18, a memory interface circuit 531, a control circuit 533, and/or a memory cell array 535, etc., but the example embodiments are not limited thereto. According to some example embodiments, one or more of the memory controller 510, the memory interface circuit 531, and/or the control circuit 533, etc., may be implemented as processing circuitry. The processing circuitry may include hardware or hardware circuit including logic circuits; a hardware/software combination such as a processor executing software and/or firmware; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc., but is not limited thereto.

[0071] The memory interface circuit 531 may receive at least one chip enable signal nCE from the

memory controller **510** through, e.g., the first pin **P11**. The memory interface circuit **531** may transmit and/or receive signals to and/or from the memory controller **510** through, e.g., the second to eighth pins **P12** to **P18**, according to and/or based on the chip enable signal **nCE**, but is not limited thereto. For example, when the chip enable signal **nCE** is in an enable state (e.g., a low level, etc.), the memory interface circuit **531** may transmit and/or receive the signals to and/or from the memory controller **510** through, e.g., the second to eighth pins **P12** to **P18**, but is not limited thereto.

[0072] The memory interface circuit **531** may receive a command latch enable signal **CLE**, an address latch enable signal **ALE**, and/or a write enable signal **nWE** from the memory controller **510** through, e.g., the second to fourth pins **P12** to **P14**, but is not limited thereto. The memory interface circuit **531** may receive a data signal **DQ** from the memory controller **510** through, e.g., the seventh pin **P17**, and/or may transmit the data signal **DQ** to the memory controller **510** through, e.g., the seventh pin **P17**, etc. A command **CMD**, an address **ADDR**, and/or data **DAT** may be transferred through the data signal **DQ**, but is not limited thereto. For example, the data signal **DQ** may be transferred through a plurality of data signal lines. In this case, the seventh pin **P17** may include a plurality of pins corresponding to a plurality of data signals, but the example embodiments are not limited thereto. For example, the data signal **DQ** may include the read command **RDCMD**, the start read address **StrRDADR**, and/or the data chunk indicator **DCNKIND**, which were described above with reference to FIG. 1 and FIG. 3, but are not limited thereto. For example, the command **CMD** may include the read command **RDCMD**, and the address **ADDR** may include the start read address **StrRDADR** and/or the data chunk indicator **DCNKIND**, etc.

[0073] The memory interface circuit **531** may obtain the command **CMD** from the data signal **DQ** received during an enable period (e.g., a high level state, etc.) of the command latch enable signal **CLE** based on toggle timings of the write enable signal **nWE**. The memory interface circuit **531** may obtain the address **ADDR** from the data signal **DQ** received during an enable period (e.g., a high level state, etc.) of the address latch enable signal **ALE** based on the toggle timings of the write enable signal **nWE**.

[0074] In at least one example embodiment, the write enable signal **nWE** may maintain a static state (e.g., a high level or a low level, etc.) and then may toggle between the high level and the low level. For example, the write enable signal **nWE** may toggle during a period in which the command **CMD** and/or the address **ADDR** is transmitted. Accordingly, the memory interface circuit **531** may acquire the command **CMD** and/or the address **ADDR** based on the toggle timings of the write enable signal **nWE**, etc.

[0075] The memory interface circuit **531** may receive a read enable signal **nRE** from the memory controller **510** through, e.g., the fifth pin **P15**. The memory interface circuit **531** may receive a data strobe signal **DQS** from the memory controller **510** through, e.g., the sixth pin **P16**, and/or may transmit the data strobe signal **DQS** to the memory controller **510** through, e.g., the sixth pin **P16**.

[0076] In an operation of outputting the data **DAT** of the nonvolatile memory device **530**, the memory interface circuit **531** may receive the read enable signal **nRE** which toggles through, e.g., the fifth pin **P15**, before the memory interface circuit **531** outputs the data **DAT**. The memory interface circuit **531** may generate the data strobe signal **DQS** which toggles based on toggling of the read enable signal **nRE**. For example, the memory interface circuit **531** may generate the data strobe signal **DQS** which starts toggling after a desired and/or predetermined delay (e.g., **tDQSRE**) based on a toggling start time point of the read enable signal **nRE**. The memory interface circuit **531** may transmit the data signal **DQ** including the data **DAT**, based on the toggle timing of the data strobe signal **DQS**. Accordingly, the data **DAT** may be aligned with the toggle timing of the data strobe signal **DQS** and transmitted to the memory controller **510**, etc.

[0077] In the data **DAT** input operation of the nonvolatile memory device **530**, when the data signal **DQ** including the data **DAT** is received from the memory controller **510**, the memory interface circuit **531** may receive the data strobe signal **DQS** toggling with the data **DAT** from the memory

controller **510**. The memory interface circuit **531** may acquire the data DAT from the data signal DQ based on the toggle timing of the data strobe signal DQS. For example, the memory interface circuit **531** may acquire the data DAT by sampling the data signal DQ at a rising edge and/or a falling edge of the data strobe signal DQS, but the example embodiments are not limited thereto. [0078] The memory interface circuit **531** may transmit a ready/busy output signal nR/B to the memory controller **510** through, e.g., the eighth pin **P18**, but is not limited thereto. The memory interface circuit **531** may transmit state information of the nonvolatile memory device **530** to the memory controller **510** through the ready/busy output signal nR/B. When the nonvolatile memory device **530** is in a busy state (e.g., when internal operations of the nonvolatile memory device **530** are being performed), the memory interface circuit **531** may transmit the ready/busy output signal nR/B indicating the busy state to the memory controller **510**. When the nonvolatile memory device **530** is in a ready state (e.g., when internal operations of the nonvolatile memory device **530** are not performed or are completed), the memory interface circuit **531** may transmit the ready/busy output signal nR/B indicating the ready state to the memory controller **510**. For example, while the nonvolatile memory device **530** reads the data DAT from the memory cell array **535** in response to the page read command, the memory interface circuit **531** may transmit the ready/busy output signal nR/B indicating the busy state (e.g., a low level, etc.) to the memory controller **510**. For example, while the nonvolatile memory device **530** programs the data DAT to the memory cell array **535** in response to a program command, the memory interface circuit **531** may transmit the ready/busy output signal nR/B indicating the busy state to the memory controller **510**, but is not limited thereto.

[0079] The control circuit **533** may control various operations of the nonvolatile memory device **530**. The control circuit **533** may receive the command/address CMD/ADDR obtained from the memory interface circuit **531**. The control circuit **533** may generate the control signals for controlling other components of the nonvolatile memory device **530** according to and/or based on the received command/address CMD/ADDR. For example, the control circuit **533** may generate various control signals for programming the data DAT in the memory cell array **535** and/or for reading the data DAT from the memory cell array **535**, etc.

[0080] The memory cell array **535** may store the data DAT obtained from the memory interface circuit **531** under the control of the control circuit **533**. The memory cell array **535** may output the stored data DAT to the memory interface circuit **531** under the control of the control circuit **533**.

[0081] The memory cell array **535** may include a plurality of memory cells. For example, the plurality of memory cells may be flash memory cells. However, the example embodiments of the inventive concepts are not limited thereto, and the memory cells may be, for example, resistive random access memory (RRAM) cells, ferroelectric random access memory (FRAM) cells, phase change random access memory (PRAM) cells, thyristor random access memory (TRAM) cells, and/or magnetic random access memory (MRAM) cells, etc.

[0082] The memory controller **510** may include a plurality of pins, e.g., first to eighth pins **P21** to **P28**, etc., and/or a controller interface circuit (CTRL I/F Circuitry) **511**, etc. The first to eighth pins **P21** to **P28** may correspond to the first to eighth pins **P11** to **P18** of the nonvolatile memory device **530**, but the example embodiments are not limited thereto.

[0083] The controller interface circuit **511** may transmit the chip enable signal nCE to the nonvolatile memory device **530** through the first pin **P21**, but is not limited thereto. The controller interface circuit **511** may transmit and/or receive the signals to and/or from the nonvolatile memory device **530** selected through the chip enable signal nCE through, e.g., the second to eighth pins **P22** to **P28**.

[0084] The controller interface circuit **511** may transmit the command latch enable signal CLE, the address latch enable signal ALE, and/or the write enable signal nWE to the nonvolatile memory device **530** through, e.g., the second to fourth pins **P22** to **P24**. The controller interface circuit **511** may transmit the data signal DQ to the nonvolatile memory device **530** through, e.g., the seventh

pin P27, and/or may receive the data signal DQ from the nonvolatile memory device 530 through, e.g., the seventh pin P27.

[0085] The controller interface circuit 511 may transmit the data signal DQ including the command CMD and/or the address ADDR to the nonvolatile memory device 530 together with the write enable signal nWE which toggles (e.g., toggles between states, etc.). The controller interface circuit 511 may transmit the data signal DQ including the command CMD to the nonvolatile memory device 530 as the controller interface circuit 511 transmits the command latch enable signal CLE having the enable state and may transmit the data signal DQ including the address ADDR to the nonvolatile memory device 530 as the controller interface circuit 511 transmits the address latch enable signal ALE having the enable state.

[0086] The controller interface circuit 511 may transmit the read enable signal nRE to the nonvolatile memory device 530 through, e.g., the fifth pin P25. The controller interface circuit 511 may receive the data strobe signal DQS from the nonvolatile memory device 530 through, e.g., the sixth pin P26, and/or may transmit the data strobe signal DQS to the nonvolatile memory device 530 through, e.g., the sixth pin P26.

[0087] In the data DAT output operation of the nonvolatile memory device 530, the controller interface circuit 511 may generate the read enable signal nRE which toggles (e.g., transitions between 0 to 1 or ON and OFF, etc.) and may transmit the read enable signal nRE to the nonvolatile memory device 530. For example, the controller interface circuit 511 may generate the read enable signal nRE which changes from a fixed state (e.g., a high level or a low level) to a toggle state (e.g., an opposite state, etc.) before the data DAT is output. Accordingly, the data strobe signal DQS which toggles based on the read enable signal nRE may be generated in the nonvolatile memory device 530. The controller interface circuit 511 may receive from the nonvolatile memory device 530, the data signal DQ including the data DAT together with the data strobe signal DQS which toggles between states. The controller interface circuit 511 may obtain the data DAT from the data signal DQ based on the toggle timing of the data strobe signal DQS.

[0088] In the data DAT input operation of the nonvolatile memory device 530, the controller interface circuit 511 may generate the data strobe signal DQS which toggles (e.g., transitions between 0 to 1 or ON and OFF, etc.). For example, the controller interface circuit 511 may generate the data strobe signal DQS which changes from a fixed state (e.g., a high level or a low level) to a toggle state before the controller interface circuit 511 transmits the data DAT, etc. The controller interface circuit 511 may transmit the data signal DQ including the data DAT to the nonvolatile memory device 530 based on the toggle timings of the data strobe signal DQS.

[0089] The controller interface circuit 511 may receive the ready/busy output signal nR/B from the nonvolatile memory device 530 through the eighth pin P28, but is not limited thereto. The controller interface circuit 511 may determine the state information of the nonvolatile memory device 530 based on the ready/busy output signal nR/B.

[0090] FIG. 7 is a block diagram illustrating at least one example embodiment of a nonvolatile memory device of FIG. 1.

[0091] Referring to FIG. 7, a nonvolatile memory device 700 may include a control circuit 710, an address decoder 730, a memory cell array 750, a page buffer circuit 770, and/or a data input/output circuit 790, etc., but is not limited thereto. According to some example embodiments, one or more of the control circuit 710, the address decoder 730, the page buffer circuit 770, and/or the data input/output circuit 790, etc., may be implemented as processing circuitry. The processing circuitry may include hardware or hardware circuit including logic circuits; a hardware/software combination such as a processor executing software and/or firmware; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc., but is not limited thereto.

[0092] The memory cell array **750** may be coupled to the address decoder **730** through at least one string selection line SSL, a plurality of word lines WLs, and/or at least one ground selection line GSL, etc. Also, the memory cell array **750** may be coupled to the page buffer circuit **770** through a plurality of bit lines BLs. The page buffer circuit **770** may be coupled to the data input/output circuit **790** through a plurality of data lines DLs.

[0093] The memory cell array **750** may include the plurality of memory cells connected to the plurality of word lines WLs and the plurality of bit lines BLs.

[0094] According to at least one example embodiment, the memory cell array **750** may be a three-dimensional memory cell array formed on a substrate in a three-dimensional structure (e.g., a vertical structure), but is not limited thereto. In this case, the memory cell array **750** may include vertical memory cell strings including a plurality of memory cells formed by being stacked on each other. A detailed description of the three-dimensional memory cell array is described in U.S. Registration Pat. Nos. 7,679,133; 8,553,466; 8,654,587; 8,559,235 and U.S. Publication No. 2011/0233648, incorporated by reference herein.

[0095] In another example embodiment, the memory cell array **750** may be a two-dimensional memory cell array formed on a substrate in a two-dimensional structure (e.g., a horizontal structure, etc.), but is not limited thereto.

[0096] The control circuit **710** may receive the chip enable signal nCE, the command latch enable signal CLE, the address latch enable signal ALE, the write enable signal nWE, and/or the read enable signal nRE described above with reference to FIG. **6**, and may output a ready/busy output signal nRB, but the example embodiments are not limited thereto.

[0097] The data input/output circuit **790** may transmit and/or receive the data signal DQ and/or the data strobe signal DQS to and/or from the memory controller. The control circuit **710** may transmit and/or receive the data signal DQ and/or the data strobe signal DQS through the data input/output circuit **790**.

[0098] The memory cell array **750** may output the first page data PGDAT1 of FIG. **1** to the page buffer circuit **770** through the plurality of bit lines BLs in a read operation of target data, and the page buffer circuit **770** may output some or all of the first page data PGDAT1 to the data input/output circuit **790** through the plurality of data lines DLs.

[0099] FIG. **8** is a timing diagram explaining at least one example embodiment of an operation of the nonvolatile memory device of FIG. **1**, **6** or **7**, but the example embodiments are not limited thereto.

[0100] In FIG. **8**, the write enable signal nWE, the read enable signal nRE, the data strobe signal DQS, and the data signal DQ are illustrated over a time period between t1 to t5. RE may be an inverted signal of the read enable signal nRE, and nDQS may be an inverted signal of the data strobe signal DQS. The signals nWE, nRE, DQS, and DQ may have states or levels as described above with reference to FIG. **6**, and the read command RDCMD, the start read address StrRDADR, and the data chunk indicator DCNKIND described above with reference to FIG. **1** and FIG. **3** may be transmitted through the data signal DQ, but the example embodiments are not limited thereto.

[0101] Referring to FIG. **1**, FIG. **7**, and FIG. **8**, the write enable signal nWE may toggle between t1 and t2, the read enable signal nRE may be maintained at the high level, and the data strobe signal DQS may have a high impedance state (e.g., “Hi-Z”). In this case, the page read command PRDCMD and the start read address StrRDADR may be received through the data signal DQ based on the toggle timings of the write enable signal nWE.

[0102] For example, the page read command PRDCMD may include a plurality of read commands, e.g., first read command PRDCMDa and a second read command PRDCMDb, but is not limited thereto. The start read address StrRDADR may include the start row address StrRA and the start column address StrCA. For example, between t1 and t2, the control circuit may sequentially receive the first read command PRDCMDa, the start read address StrRDADR, and the second read command PRDCMDb, but is not limited thereto. For example, when the memory cells in the

memory cell array include multi-level cells, before the first read command PRDCMDa is received, a page selection command PSLTCMD indicating one of least significant bit (LSB) page, a central significant bit (CSB) page, and a most significant bit (MSB) page may be further received, etc. [0103] Between t2 and t3, the write enable signal nWE may maintain the high level, the read enable signal nRE may maintain the low level, and the data strobe signal DQS may maintain the high impedance state, but are not limited thereto. For example, between t2 and t3, during a time interval tR, the memory cell array may output page data (e.g., PGDAT1 of FIG. 1) to the page buffer circuit, etc.

[0104] Between t3 and t4, the write enable signal nWE may toggle again and may have the high level, the read enable signal nRE may first maintain the high level and then toggle, and the data strobe signal DQS may maintain the high impedance state. In this case, data read commands DRDCMDa and DRDCMDb and the data chunk indicator DCNKIND may be received through the data signal DQ based on the toggle timings of the write enable signal nWE, but the example embodiments are not limited.

[0105] For example, the data read command DRDCMD may include the third read command DRDCMDa and the fourth read command DRDCMDb, but is not limited thereto. The data chunk indicator DCNKIND may individually indicate whether the sizes of each of the data chunks are partial (e.g., has a size less than a whole page of data) or entire of the page data (e.g., are equal to the size of a whole page of data) output to the page buffer circuit between t2 and t3. For example, between t3 and t4, the control circuit may sequentially receive the third read command DRDCMDa, the data chunk indicator DCNKIND, and the fourth read command DRDCMDb, but is not limited thereto.

[0106] For example, the read enable signal nRE may toggle after a desired and/or predetermined delay (e.g., tWHR2) from the time when the write enable signal nWE has the high level after toggling.

[0107] Between t4 and t5, the write enable signal nWE may maintain the high level, and the read enable signal nRE and the data strobe signal DQS may toggle, but is not limited thereto. In this case, the output data ODAT1x may be output through the data signal DQ based on the toggle timings of the data strobe signal DQS.

[0108] For example, the output data ODAT1x may be output after a desired and/or predetermined delay (e.g., tDQSRE) based on a toggling start time (e.g., between t3 and t4, etc.) of the read enable signal nRE. For example, as described above with reference to FIG. 1, the output data ODAT1x may include one or more of the plurality of first data chunks, and the page buffer circuit may output and/or continuously output two or more of the plurality of first data chunks after a desired and/or predetermined time (e.g., tWHR2+tDQSRE) elapses from a time point at which the nonvolatile memory device receives the fourth read command DRDCMDb, but the example embodiments are not limited thereto.

[0109] In at least one example embodiment, the control circuit may sequentially receive the first read command PRDCMDa, the start read address StrRDADR, the second read command PRDCMDb, the third read command DRDCMDa, the data chunk indicator DCNKIND, and the fourth read command DRDCMDb between t1 and t4, but the example embodiments are not limited thereto.

[0110] FIG. 9 is a flowchart explaining an operation in which a control circuit of FIG. 7 receives a read command, a start read address, and a data chunk indicator according to at least one example embodiment.

[0111] Referring to FIGS. 1, 5, 7, 8, and 9, the control circuit (e.g., 310 of FIG. 1, 533 of FIG. 6, 710 of FIG. 7, etc.) may receive the page read command PRDCMD and the start read address StrRDADR (S110), but the example embodiments are not limited thereto.

[0112] In at least one example embodiment, the page read command PRDCMD may be a command for outputting page data stored in one page of the memory cell array (e.g., 330 of FIG. 1, 535 of

FIG. 6, 750 of FIG. 7, etc.) from the memory cell array to the page buffer circuit (e.g., 350 of FIG. 1, 770 of FIG. 7, etc.).

[0113] The control circuit may receive the data read command DRDCMD and the data chunk indicator DCNKIND, etc., (S130).

[0114] In at least one example embodiment, the data read command DRDCMD may be a command for outputting some or all of the page data from the page buffer circuit to the data input/output circuit (e.g., 790 of FIG. 7, etc.).

[0115] FIG. 10 is a flowchart illustrating at least one example embodiment of an operation in which a control circuit of FIG. 7 receives a read command and a start read address.

[0116] Referring to FIG. 1, FIG. 5, FIG. 7, FIG. 8, and FIG. 9, the control circuit (e.g., 310 of FIG. 1, 533 of FIG. 6, 710 of FIG. 7, etc.) may sequentially receive the first read command PRDCMDa, the start read address StrRDADR, and the second read command PRDCMDb (S111), but the example embodiments are not limited thereto.

[0117] In at least one example embodiment, the page read command PRDCMD may include the first read command PRDCMDa and the second read command PRDCMDb, etc.

[0118] According to at least one example embodiment, S110 of FIG. 9 may include S111 of FIG. 10, but the example embodiments are not limited thereto.

[0119] FIG. 11 is a flowchart illustrating at least one example embodiment of an operation in which a control circuit of FIG. 7 receives a read command and a data chunk indicator.

[0120] Referring to FIG. 1, FIG. 5, FIG. 7, FIG. 8, and FIG. 9, the control circuit (e.g., 310 of FIG. 1, 533 of FIG. 6, 710 of FIG. 7, etc.) may sequentially receive the third read command DRDCMDa, the data chunk indicator DCNKIND, and the fourth read command DRDCMDb, etc., (S131), but is not limited thereto.

[0121] In at least one example embodiment, the data read command DRDCMD may include the third read command DRDCMDa and the fourth read command DRDCMDb, but the example embodiments are not limited thereto.

[0122] In at least one example embodiment, S130 of FIG. 9 may include S131 of FIG. 11, but the example embodiments are not limited thereto.

[0123] FIG. 12A and FIG. 12B are diagrams illustrating at least one example embodiment of a data chunk indicator of FIG. 1.

[0124] In FIG. 12A and FIG. 12B, first page data PGDAT1y and data chunk indicators DCNKINDa and DCNKINDb are illustrated, but the example embodiments are not limited thereto. The first page data PGDAT1y may correspond to the page data (e.g., PGDAT1 of FIG. 1, PGDAT1x of FIG. 3, etc.) output from the memory cell array to the page buffer circuit.

[0125] As described above with reference to FIG. 1, the data chunk indicator DCNKIND may individually indicate a plurality of first data chunks, e.g., DCNK11y, DCNK12y, DCNK13y, DCNK14y, DCNK15y, DCNK16y, DCNK17y, and DCNK18y, etc., included in the first page data PGDAT1y output from the memory cell array (e.g., 330 of FIG. 1, etc.). Data chunks indicated by the data chunk indicator DCNKIND may be output from the page buffer circuit (e.g., 350 of FIG. 1, etc.) to the data input/output circuit.

[0126] Referring to FIG. 1 and FIG. 12A, the data chunk indicator DCNKINDa may include binary data indicating one or more of the plurality of first data chunks DCNK11y to DCNK18y, but is not limited thereto.

[0127] In at least one example embodiment, the data chunk indicator DCNKINDa may indicate each of the plurality of first data chunks DCNK11y to DCNK18y using a bitmap type, but is not limited thereto.

[0128] For example, the number of plurality of first data chunks DCNK11y to DCNK18y may be N (N being an integer of 2 or more), and in this case, the data chunk indicator DCNKINDa may be N-bit data.

[0129] For example, when the data chunk indicator DCNKINDa indicates the data chunk

DCNK18y, the data chunk indicator DCNKINDa may have a value of “00000001”. When the data chunk indicator DCNKINDa indicates the data chunk DCNK17y, the data chunk indicator DCNKINDa may have a value of “00000010”. When the data chunk indicator DCNKINDa indicates the data chunks DCNK17y and DCNK18y, the data chunk indicator DCNKINDa may have a value of “00000011”. When the data chunk indicator DCNKINDa indicates the data chunks DCNK13y, DCNK14y, DCNK15y, and DCNK18y, the data chunk indicator DCNKINDa may have a value of “00111001”. When the data chunk indicator DCNKINDa indicates the data chunks DCNK11y to DCNK13y, DCNK15y, and DCNK18y, the data chunk indicator DCNKINDa may have a value of “11101001”. When the data chunk indicator DCNKINDa indicates the data chunks DCNK11y to DCNK18y, the data chunk indicator DCNKINDa may have a value of “11111111”. However, the example embodiments are not limited thereto, and the data chunk indicators may have other values.

[0130] Referring to FIG. 1 and FIG. 12B, the data chunk indicator DCNKINDb may include decimal data indicating one or more of the plurality of data chunks DCNK11y to DCNK18y, etc.

[0131] In at least one example embodiment, the data chunk indicator DCNKINDb may indicate each of the plurality of data chunks DCNK11y to DCNK18y by using an offset type. For example, a data chunk having a smallest column address from among the plurality of data chunks DCNK11y to DCNK18y may be a reference, but the example embodiments are not limited thereto. The remaining data chunks excluding the data chunk as a reference may be expressed based on a distance from the reference data chunk.

[0132] For example, the number of the plurality of first data chunks DCNK11y to DCNK18y may be N, and in this case, the data chunk indicator DCNKINDb may be data having a digit less than or equal to N.

[0133] For example, when the data chunk indicator DCNKINDb indicates the data chunk DCNK18y, the data chunk indicator DCNKINDb may have a value of “8”. When the data chunk indicator DCNKINDb indicates the data chunk DCNK17y, the data chunk indicator DCNKINDb may have a value of “7”. When the data chunk indicator DCNKINDb indicates the data chunks DCNK17y and DCNK18y, the data chunk indicator DCNKINDb may have values of “7” and “8”. When the data chunk indicator DCNKINDb indicates the data chunks DCNK13y, DCNK14y, DCNK15y, and DCNK18y, the data chunk indicator DCNKINDb may have values of “3”, “4”, “5”, and “8”. When the data chunk indicator DCNKINDb indicates the data chunks DCNK11y to DCNK13y, DCNK15y, and DCNK18y, the data chunk indicator DCNKINDb may have values of “1”, “2”, “3”, “5”, and “8”. When the data chunk indicator DCNKINDb indicates the data chunks DCNK11y to DCNK18y, the data chunk indicator DCNKINDb may have values of “1”, “2”, “3”, “4”, “5”, “6”, “7”, and “8”.

[0134] With reference to FIG. 12A and FIG. 12B, some example embodiments in which the data chunk indicator DCNKIND includes binary data or decimal data are described, but the scope of the example embodiments of the inventive concepts are not limited thereto. In another example embodiment, the data chunk indicator DCNKIND may have various data formats which may respectively indicate one or more of the plurality of data chunks DCNK11y to DCNK18y.

[0135] FIG. 13 is a flowchart illustrating at least one example embodiment of an operation in which a control circuit of FIG. 7 outputs one or more of a plurality of first data chunks.

[0136] Referring to FIG. 1, FIG. 5, FIG. 7, FIG. 8, and FIG. 13, the page buffer circuit (e.g., 350 in FIG. 1, 770 in FIG. 7, etc.) may output and/or continuously output two or more data chunk indicators (e.g., DCNKa, . . .) indicated by the data chunk indicator DCNKIND from among the plurality of first data chunks DCNK11, DCNK12, DCNK13, and the like (S510).

[0137] In at least one example embodiment, S500 of FIG. 5 may include S510 of FIG. 13, but is not limited thereto.

[0138] FIG. 14 is a timing diagram explaining at least one example embodiment of an operation of a nonvolatile memory device of FIG. 7. As described above with reference to FIG. 2, data chunks

may be defined not for one page but for data sizes smaller than one page, such as one block or one plane, etc. When the sizes of data chunks are defined as being equal to one block, the data chunks may be referred to as “a plurality of first block data chunks”, and when the sizes of data chunks are defined as being equal to one plane, the data chunks may be referred to as “a plurality of first plane data chunks”, etc. Each of the plurality of first block data chunks or the plurality of first plane data chunks may be processed in the same as, or similar manner, to the plurality of first data chunks DCNK11, DCNK12, DCNK13, and the like. Thus, the page buffer circuit (e.g., 350 in FIG. 1, etc.) may output one or more of the plurality of first block data chunks defined as being equal to the size of one block based on the data chunk indicator or one or more of the plurality of first plane data chunks defined as being equal to the size of one plane, and may output and/or continuously output two or more of the plurality of first block data chunks or two or more of the plurality of first plane data chunks, etc., but the example embodiments are not limited thereto.

[0139] In at least one example embodiment, each of the plurality of first block data chunks or the plurality of first plane data chunks may be stored in memory cells arranged in the same row of the memory cell array as the data chunks described above with reference to FIG. 8, and may be stored in memory cells arranged in two or more different rows of the memory cell array, unlike the above data chunks, but the example embodiments are not limited thereto. Thus, in a manner similar to the manner described above with reference to FIG. 3, to indicate the start of the plurality of first block data chunks or the start of the plurality of first plane data chunks, the number of read commands included in the page read command (e.g., PRDCMD in FIG. 1) may remain unchanged (e.g., two) or may increase (e.g., three or more), and the number of start row addresses and the number of start column addresses included in the start read address (e.g., StrRDADR in FIG. 1) may also remain unchanged (e.g., one, respectively) or may increase (e.g., two or more, respectively). However, the number of read commands included in the data read command (e.g., DRDCMD in FIG. 1) may remain unchanged (e.g., two) regardless of whether the plurality of first block data chunks or the plurality of first plane data chunks are stored in memory cells placed in the same row of the memory cell array or placed in two or more different rows of the memory cell array. The number of data chunk indicators for indicating one or more of the plurality of first block data chunks or one or more of the plurality of first plane data chunks may be unchanged or may be increased compared to the case indicating a plurality of data chunks.

[0140] For example, when each of the plurality of first block data chunks or each of the plurality of first plane data chunks is stored in memory cells arranged in the same row of the memory cell array, in the same manner as described above with reference to FIG. 8, the page read command may include two read commands, the start read address may include one start row address and one start column address, and the data read command may include two read commands, but the example embodiments are not limited thereto. The number of data chunk indicators may be one, but is not limited thereto.

[0141] For example, when each of the plurality of first block data chunks or each of the plurality of first plane data chunks are stored in memory cells arranged in two different rows of the memory cell array, as illustrated in FIG. 14, the page read command may include four read commands, the start read address may include two start row addresses and two start column addresses, and the data read command may include two read commands, but the example embodiments are not limited thereto. The number of data chunk indicators may be two, but is not limited thereto.

[0142] Referring to FIG. 14, when each of the plurality of first block data chunks or each of the plurality of first plane data chunks are stored in memory cells arranged in two different rows of a memory cell array, at least one example embodiment of outputting one or more of the plurality of first block data chunks or one or more of the plurality of plane data chunks is illustrated, but the example embodiments are not limited thereto.

[0143] In at least one example embodiment, the control circuit (e.g., 310 of FIG. 1, 533 of FIG. 6, and 710 of FIG. 7) may receive the page read command including, e.g., read commands

PRDCMD-1, PRDCMD-2, PRDCMD-3, and PRDCMD-4, etc., and a data read command including, e.g., read commands DRDCMD-1 and DRDCMD-2, etc., but the example embodiments are not limited thereto. The control circuit may receive a first start read address including a first start row address StrRA1 and a first start column address StrCA1, and a second start read address including a second start row address StrRA2 and a second start column address StrCA2, etc. The control circuit may receive a first data chunk indicator DCNKIND1 and a second data chunk indicator DCNKIND2 indicating one or more of the plurality of first block data chunks or one or more of the plurality of plane data chunks.

[0144] For example, the control circuit may sequentially receive the read command PRDCMD-1, the first start column address StrCA1, the first start row address StrRA1, and the read command PRDCMD-2, but the example embodiments are not limited thereto. The control circuit may sequentially receive the read command PRDCMD-3, the second start column address StrCA2, the second start row address StrRA2, and the read command PRDCMD-4, etc. During the tR time period, the memory cell array (e.g., 330 in FIG. 1, etc.) may output block data including the plurality of first block data chunks or plane data including the plurality of plane data chunks to the page buffer circuit (e.g., 350 in FIG. 1, etc.) based on the first start row address StrRA1 and the second start row address StrRA2.

[0145] For example, the control circuit may sequentially receive the read command DRDCMD-1, the first data chunk indicator DCNKIND1, the second data chunk indicator DCNKIND2, and the read command DRDCMD-2, etc. The first data chunk indicator DCNKIND1 may indicate one or more of the data block chunks or the data plane chunks stored in memory cells arranged in one row, etc., and the second data chunk indicator DCNKIND2 may indicate one or more of the data block chunks or the data plane chunks stored in memory cells arranged in another row, etc.

[0146] For example, the page buffer circuit may output and/or continuously output two or more of the plurality of first block data chunks and/or two or more of the plurality of first data page chunks based on the first data chunk indicator DCNKIND1 and the second data chunk indicator DCNKIND2, but is not limited thereto.

[0147] In at least one example embodiment, when the memory cells of the memory cell array include multi-level cells, the control circuit may further receive the page selection command PSLTCMD before the control circuit receives the read command PRDCMD-1, but the example embodiments are not limited thereto.

[0148] FIG. 15 is a block diagram illustrating at least one example embodiment of a memory controller communicating with a nonvolatile memory device of FIG. 1, but the example embodiments are not limited thereto.

[0149] Referring to FIG. 1 and FIG. 15, a memory controller 900 may correspond to the memory controller 100 of FIG. 1, but is not limited thereto.

[0150] The memory controller 900 may include at least one processor 910, a garbage collection circuit 920, an address mapping table 930, a host interface 940, an NVM interface 950, and/or at least one bus 960, etc., but is not limited thereto. The garbage collection circuit 920 may manage a plurality of read address ranges RDADDRNGs 921 described above with reference to FIG. 1 and FIG. 4, but the example embodiments are not limited thereto. According to some example embodiments, one or more of the at least one processor 910, the garbage collection circuit 920, the address mapping table 930, the host interface 940, the NVM interface 950, and/or the at least one bus 960, etc., may be implemented as processing circuitry. The processing circuitry may include hardware or hardware circuit including logic circuits; a hardware/software combination such as a processor executing software and/or firmware; or a combination thereof. For example, the processing circuitry more specifically may include, but is not limited to, a central processing unit (CPU), an arithmetic logic unit (ALU), a digital signal processor, a microcomputer, a field programmable gate array (FPGA), a System-on-Chip (SoC), a programmable logic unit, a microprocessor, application-specific integrated circuit (ASIC), etc., but is not limited thereto.

[0151] The processor **910** may generally control the components **920, 930, 940, 950**, and/or **960** of the memory controller **900**, but is not limited thereto. The bus **960** may perform communication between the components **910, 920, 930, 940**, and/or **950** of the memory controller **900**, etc., but is not limited thereto.

[0152] The garbage collection circuit **920** may perform the garbage collection on the memory cell array (e.g., **330** of FIG. **1**) of the nonvolatile memory device (e.g., **300** of FIG. **1**) by using the address mapping table **930**. For example, the garbage collection may be performed to secure and/or generate free blocks in the memory cell array by deallocating erased blocks in the memory cell array. In the process of performing the garbage collection, the garbage collection circuit **920** may generate address ranges for accessing memory cells targeted for garbage collection. The generated address ranges may be managed as a plurality of read address ranges **921**. In this case, the plurality of read address ranges **921** may be continuous or discontinuous in an address space of the memory cell array. When the plurality of read address ranges **921** are continuous in the process of performing the garbage collection, data chunks output from the page buffer circuit of the nonvolatile memory device may be continuous in latches of the page buffer circuit, and when the plurality of read address ranges **921** are discontinuous, they may be discontinuous in latches of the page buffer circuit. In this case, according to some example embodiments of the inventive concepts described above with reference to FIG. **1** to FIG. **14**, the nonvolatile memory device may output and/or continuously output a plurality of target data in units of data chunk based on the data chunk indicator.

[0153] In at least one example embodiment, the memory controller **900** may output the read command RDCMD, the start read address StrRDADR, and/or the data chunk indicator DCNKIND described above with reference to FIG. **1** to the nonvolatile memory device through the NVM interface **950**, but the example embodiments are not limited thereto.

[0154] In other example embodiments, the read command RDCMD, the start read address StrRDADR, and/or the data chunk indicator DCNKIND may be transmitted from an external host device via the host interface **940**. For example, the memory controller **900** may further include an error correction code (ECC) circuit which performs error correction on data to enhance and/or increase the reliability of data stored in the nonvolatile memory device. The ECC circuit may generate parity bits for data, and the memory controller **900** may store the parity bits in separate regions of the nonvolatile memory device. In this case, the plurality of read address ranges **921** associated with the parity bits may be discontinuous similar to the case described above with reference to FIG. **15**, and according to some example embodiments of the inventive concepts, the nonvolatile memory device may output and/or continuously output a plurality of target data in units of data chunk based on a data chunk indicator.

[0155] FIG. **16** is a flowchart illustrating an operating method of a nonvolatile memory device according to at least one example embodiment of the inventive concepts.

[0156] Referring to FIG. **1**, FIG. **5**, and FIG. **16**, the control circuit **310** may receive the read command RDCMD, the start read address StrRDADR, and/or the data chunk indicator DCNKIND (**S100**). The memory cell array **330** may output the first page data PGDAT1 including the plurality of first data chunks, e.g., DCNK11, DCNK12, DCNK13, and the like, based on the start row address StrRA (**S300**). The page buffer circuit **350** may output one or more (e.g., DCNKa, . . .) of the plurality of first data chunks DCNK11, DCNK12, DCNK13, and the like, based on the data chunk indicator DCNKIND (**S500**).

[0157] Before operation **S100** is performed, the control circuit **310** may receive a chunk size indicator which indicates a size of the data chunk which may be some or all of page data and is a unit of a set of data output from the page data circuit (**S50**).

[0158] In at least one example embodiment, before the control circuit **310** receives a read command (e.g., RDCMD in FIG. **1**), the control circuit **310** may receive the chunk size indicator indicating the size of each of the plurality of first data chunks (e.g., DCNK11, DCNK12, DCNK13, and the

like in FIG. 1).

[0159] For example, operation **S50** may be an optional operation, e.g., may not be performed every time operation **S100** is performed, and for example, may be performed one-time or may be performed less times than the number of times **S100** is performed. For example, the control circuit **310** may receive the chunk size indicator at the time of shipping, initializing, and/or powering on the nonvolatile memory device (e.g., **300** of FIG. 1, **530** of FIG. 6, **700** of FIG. 7, etc.). The chunk size indicator may be received from an external host device and/or a memory controller controlling the nonvolatile memory device, and may be temporarily and/or non-temporarily stored in the nonvolatile memory device and/or the memory controller. For example, when the chunk size indicator is stored in the nonvolatile memory device, the chunk size indicator may be stored in a separate area within the control circuit and/or the memory cell array of the nonvolatile memory device, and when the chunk size indicator is stored in the memory controller, it may be stored in a buffer memory in the memory controller, but the example embodiments are not limited thereto.

[0160] As described above with reference to FIG. 1, one data chunk may be a set of data defined by dividing page data evenly or unevenly (e.g., evenly dividing the data size of a page of data, or unevenly dividing the data size of a page of data), and the chunk size indicator may be transmitted from the memory controller to the nonvolatile memory device before **S100**. Regardless of whether the sizes of data chunks included in one page data are equal or not, some example embodiments of the inventive concepts may be performed based on the data chunk indicator **DCNKIND** described above with reference to FIG. 1, FIG. 12A, and FIG. 12B, but is not limited thereto.

[0161] FIG. 17 is a diagram illustrating a data center including the memory system according to at least one example embodiment of the inventive concepts.

[0162] Referring to FIG. 17, the data center **3000** may be a facility that collects various types and/or pieces of data and/or provides services, and may be referred to as a data storage center, etc. For example, the data center **3000** may be a system for operating a search engine and/or a database, and may be a computing system used by companies, such as banks, etc., and/or government agencies, etc. The data center **3000** may include application servers **3100** to **3100n** and storage servers **3200** to **3200m**. The number of application servers **3100** to **3100n** and the number of storage servers **3200** to **3200m** may vary. The number of application servers **3100** to **3100n** may be different from the number of storage servers **3200** to **3200m**.

[0163] The application server **3100** and/or the storage server **3200** may include at least one of processors **3110** and **3210** and memories **3120** and **3220**, etc., but the example embodiments are not limited thereto. The storage server **3200** will now be described as an example. The processor **3210** (e.g., processing circuitry, etc.) may control all operations of the storage server **3200**, access the memory **3220**, and/or execute instructions and/or data loaded in the memory **3220**, etc., but is not limited thereto. The memory **3220** may be, e.g., a double-data-rate synchronous DRAM (DDR SDRAM), a high-bandwidth memory (HBM), a hybrid memory cube (HMC), a dual in-line memory module (DIMM), Optane DIMM, and/or a non-volatile DIMM (NVMDIMM), but the example embodiments are not limited thereto. In some example embodiments, the numbers of processors **3210** and/or memories **3220** included in the storage server **3200** may be variously selected. In at least one example embodiment, the processor **3210** and the memory **3220** may provide a processor-memory pair. In at least one example embodiment, the number of processors **3210** may be different from the number of memories **3220**. The processor **3210** may be one or more processors, and may include a single-core processor and/or a multi-core processor. The above description of the storage server **3200** may be similarly applied to the application server **3100**. In some example embodiments, the application server **3100** may not include a storage device **3150**. The storage server **3200** may include at least one storage device **3250**. The number of storage devices **3250** included in the storage server **3200** may be variously selected according to some example embodiments.

[0164] The application servers **3100** to **3100n** may communicate with the storage servers **3200** to

3200m through at least one network **3300**. The network **3300** may be a wired network implemented using, e.g., a fiber channel (FC) and/or Ethernet, etc., but is not limited thereto, and/or the network **3300** may be a wireless network, such as a WiFi network, a cellular network (e.g., 4G LTE network, 5G NR network, a 6G network, etc.), a satellite network, etc., but the example embodiments are not limited thereto. In the case where the network **3300** is implemented using FC, the FC may be a medium used for relatively high-speed data transmission and may use an optical switch with high performance and/or high availability, etc., but the example embodiments are not limited thereto. The storage servers **3200** to **3200m** may be provided as file storages, block storages, and/or object storages according to an access method of the network **3300**, etc.

[0165] In at least one example embodiment, the network **3300** may be a storage-dedicated network, such as a storage area network (SAN), etc. For example, the SAN may be an FC-SAN, which uses an FC network and is implemented according to an FC protocol (FCP), but is not limited thereto. As another example, the SAN may be an Internet protocol (IP)-SAN, which uses a transmission control protocol (TCP)/IP network and is implemented according to a SCSI over TCP/IP or Internet SCSI (iSCSI) protocol, but the example embodiments are not limited thereto. In another example embodiment, the network **3300** may be a general network, such as a TCP/IP network, etc. For example, the network **3300** may be implemented according to a protocol, such as FC over Ethernet (FCOE), network attached storage (NAS), and/or NVMe over Fabrics (NVMe-oF), etc.

[0166] Hereinafter, the application server **3100** and the storage server **3200** will mainly be described. A description of the application server **3100** may be applied to another application server **3100n**, and/or a description of the storage server **3200** may be applied to another storage server **3200m**, etc.

[0167] The application server **3100** may store data, which was requested by a user and/or at least one client, in one or more of the storage servers **3200** to **3200m** through the network **3300**. Also, the application server **3100** may obtain data, which was requested by the user and/or the client to be read from one or more of the storage servers **3200** to **3200m** through the network **3300**. For example, the application server **3100** may be implemented as a web server, a file server, and/or a database management system (DBMS), etc.

[0168] The application server **3100** may access a memory **3120n** and/or a storage device **3150n**, which is included in another application server **3100n**, through the network **3300**. Additionally or alternatively, the application server **3100** may access memories **3220** to **3220m** and/or storage devices **3250** to **3250m**, which are included in the storage servers **3200** to **3200m**, through the network **3300**. Thus, the application server **3100** may perform various operations on data stored in application servers **3100** to **3100n** and/or the storage servers **3200** to **3200m**. For example, the application server **3100** may execute an instruction for moving and/or copying data between the application servers **3100** to **3100n** and/or the storage servers **3200** to **3200m**. In this case, the data may be moved from the storage devices **3250** to **3250m** of the storage servers **3200** to **3200m** to the memories **3120** to **3120n** of the application servers **3100** to **3100n** directly and/or through the memories **3220** to **3220m** of the storage servers **3200** to **3200m**. The data moved through the network **3300** may be data encrypted for security and/or privacy, etc.

[0169] The storage server **3200** will now be described as an example. An interface **3254** may provide physical connection between at least one processor **3210** and at least one controller **3251** and a physical connection between at least one network interface card (NIC) **3240** and the controller **3251**. For example, the interface **3254** may be implemented using, e.g., a direct attached storage (DAS) scheme in which the storage device **3250** is directly connected with at least one dedicated cable, etc. For example, the interface **3254** may be implemented by using various interface schemes, such as ATA, SATA, e-SATA, an SCSI, SAS, PCI, PCIe, NVMe, IEEE 1394, a USB interface, an SD card interface, an MMC interface, an eMMC interface, a UFS interface, an eUFS interface, and/or a CF card interface, but is not limited thereto.

[0170] The storage server **3200** may further include at least one switch **3230** and/or the NIC

(Network InterConnect) **3240**, etc. The switch **3230** may selectively connect the processor **3210** to the storage device **3250** and/or selectively connect the NIC **3240** to the storage device **3250** via the control of the processor **3210**.

[0171] In at least one example embodiment, the NIC **3240** may include at least one network interface card and/or at least one network adaptor, etc. The NIC **3240** may be connected to the network **3300** by a wired interface, a wireless interface, a Bluetooth interface, and/or an optical interface, etc., but is not limited thereto. The NIC **3240** may include an internal memory, a digital signal processor (DSP), and/or a host bus interface and be connected to the processor **3210** and/or the switch **3230** through the host bus interface. The host bus interface may be implemented as one of the above-described examples of the interface **3254**, but is not limited thereto. In at least one example embodiment, the NIC **3240** may be integrated with at least one of the processor **3210**, the switch **3230**, and/or the storage device **3250**, etc.

[0172] In the storage servers **3200** to **3200m** and/or the application servers **3100** to **3100n**, at least one processor may transmit at least one command to storage devices **3150** to **3150n** and/or **3250** to **3250m** and/or the memories **3120** to **3120n** and/or **3220** to **3220m** to program and/or read data, etc. In this case, the data may be data of which an error is corrected by an ECC engine, but is not limited thereto. The data may be data on which a data bus inversion (DBI) operation and/or a data masking (DM) operation is performed, and may include cyclic redundancy code (CRC) information, but the example embodiments are not limited thereto. The data may be data encrypted for security and/or privacy, etc.

[0173] Storage devices **3150** to **3150n** and **3250** to **3250m** may transmit at least one control signal and/or a command/address signal to NAND flash memory devices **3252** to **3252m** in response to at least one read command received from the processor, but is not limited thereto. Thus, when data is read from the NAND flash memory devices **3252** to **3252m**, a read enable (RE) signal may be input as a data output control signal, and thus, the data may be output to a DQ bus. A data strobe signal DQS may be generated using the RE signal. The command and the address signal may be latched in a page buffer depending on a rising edge and/or falling edge of a write enable (WE) signal.

[0174] The controller **3251** may control all operations of the storage device **3250**. In at least one example embodiment, the controller **3251** may include SRAM, but is not limited thereto. The controller **3251** may write data to the NAND flash memory device **3252** in response to a write command and/or read data from the NAND flash memory device **3252** in response to a read command, etc. For example, the write command and/or the read command may be provided from the processor **3210** of the storage server **3200**, the processor **3210m** of another storage server **3200m**, and/or the processors **3110** and **3110n** of the application servers **3100** and **3100n**. DRAM **3253** may temporarily store (and/or buffer) data to be written to the NAND flash memory device **3252** and/or data read from the NAND flash memory device **3252**. Also, the DRAM **3253** may store metadata, etc. Here, the metadata may be user data and/or data generated by the controller **3251** to manage the NAND flash memory device **3252**, but is not limited thereto. The storage device **3250** may include a secure element (SE) for security and/or privacy, etc. Some or all of the storage devices **3150** to **3150n** and **3250** to **3250m** may be a memory system according to some example embodiments of the inventive concepts. Some or all of the application servers **3100** to **3100n** may be the host devices referred to in the process of describing the memory system according to one or more example embodiments of the inventive concepts. However, the example embodiments of the inventive concepts are not limited thereto. Accordingly, some or all of the storage devices **3150** to **3150n** and **3250** to **3250m** may efficiently process at least one access request from a first application processor among the application servers **3100** to **3100n** and at least one key change request from a second application processor among the application servers **3100** to **3100n**, etc., but are not limited thereto.

[0175] Each of the storage devices **3250** to **3250m** may include a non-volatile memory device according to one or more of the example embodiments of the inventive concepts. The NAND flash

memory devices 3252 to 3252m may correspond to the non-volatile memory devices according to one or more of the example embodiments of the inventive concepts described above with reference to FIG. 1 to FIG. 16. Thus, the NAND flash memory devices 3252 to 3252m may improve the input/output performance of the NAND flash memory devices 2352 to 2352m by decreasing and/or preventing intervention of additional signals.

[0176] As described above, the nonvolatile memory device according to one or more of the example embodiments of the inventive concepts may output and/or continuously output a plurality of target data targeted for a data read operation in units of data chunks based on the data chunk indicator, and the target data may be a part of or all of the page data. In other words, the maximum size of a data chunk may be less than or equal to a maximum size of a page of data. Thus, additional signals (e.g., additional commands and/or additional addresses, etc.) need not be generated, issued, intervened (and/or applied) between one data chunk output from the nonvolatile memory device and another data chunk. The nonvolatile memory device may decrease, prevent, and/or remove intervention of the additional signals, and thus the input/output performance of the nonvolatile memory device and/or a memory system may be improved.

[0177] The above description is of specific example embodiments of the inventive concepts. However, the inventive concepts will include not only the above-described example embodiments, but also example embodiments changes or modifications thereto. In addition, the example embodiments of the inventive concepts will also include technologies which may be easily modified and implemented by using one or more example embodiments. Therefore, the scope of the inventive concepts should not be limited to the above-described example embodiments, but should be determined not only by the scope of the claims to be described later but also by those equivalent to the claims of the inventive concepts.

[0178] While various example embodiments of the inventive concepts have been described, it will be apparent to those of ordinary skill in the art that various changes and modifications may be made thereto without departing from the spirit and scope of the inventive concepts as set forth in the following claims.

Claims

1. A nonvolatile memory device comprising: processing circuitry configured to receive a read command, a start read address, and a data chunk indicator, the read command including a page read command and a data read command, and the start read address including a start row address and a start column address; a memory cell array configured to output first page data based on the start row address, the first page data including a plurality of first data chunks; and the processing circuitry further configured to output one or more of the plurality of first data chunks based on the data chunk indicator.
2. The nonvolatile memory device of claim 1, wherein the plurality of first data chunks correspond to a plurality of read address ranges, respectively; and the processing circuitry is further configured to continuously output two or more of the plurality of first data chunks.
3. The nonvolatile memory device of claim 2, wherein the plurality of read address ranges include two or more discontinuous address ranges of the memory cell array.
4. The nonvolatile memory device of claim 1, wherein the data chunk indicator includes binary data indicating one or more of the plurality of first data chunks.
5. The nonvolatile memory device of claim 4, wherein a number of the plurality of first data chunks is N, wherein N is an integer of 2 or more; and the data chunk indicator has a size of N-bits.
6. The nonvolatile memory device of claim 1, wherein the processing circuitry is further configured to: receive the page read command and the start read address; and receive the data read command and the data chunk indicator.
7. The nonvolatile memory device of claim 6, wherein the page read command includes a first read

- command and a second read command; and the data read command includes a third read command and a fourth read command.
- 8.** The nonvolatile memory device of claim 7, wherein the processing circuitry is further configured to: sequentially receive the third read command, the data chunk indicator, and the fourth read command.
- 9.** The nonvolatile memory device of claim 7, wherein the processing circuitry is further configured to: sequentially receive the first read command, the start read address, the second read command, the third read command, the data chunk indicator, and the fourth read command; and in response to a desired time period elapsing from a time the nonvolatile memory device received the fourth read command, sequentially output two or more of the plurality of first data chunks.
- 10.** The nonvolatile memory device of claim 1, wherein, before the read command is received, the processing circuitry is further configured to: receive a chunk size indicator indicating a size of each of the plurality of first data chunks.
- 11.** The nonvolatile memory device of claim 10, wherein, in response to the nonvolatile memory device being initialized or powered on, the processing circuitry is further configured to: receive the chunk size indicator; and store the chunk size indicator.
- 12.** The nonvolatile memory device of claim 1, wherein the processing circuitry is further configured to: receive a read enable signal which toggles based on a size of each of the plurality of first data chunks; and generate a data strobe signal which toggles based on the read enable signal.
- 13.** The nonvolatile memory device of claim 12, wherein the processing circuitry is further configured to: output one or more of the plurality of first data chunks based on the data chunk indicator and the data strobe signal.
- 14.** The nonvolatile memory device of claim 1, wherein the memory cell array is configured to output block data or plane data based on the start row address, the block data including a plurality of first block data chunks, and the plane data including a plurality of first plane data chunks; and the processing circuitry is further configured to sequentially output two or more of the plurality of first block data chunks or two or more of the plurality of first plane data chunks based on the data chunk indicator.
- 15.** A method of operating a nonvolatile memory device, the method comprising: receiving a read command, a start read address, and a data chunk indicator, the read command including a page read command and a data read command, and the start read address including a start row address and a start column address; outputting first page data based on the start row address, the first page data including a plurality of first data chunks; and outputting one or more of the plurality of first data chunks based on the data chunk indicator.
- 16.** The method of claim 15, wherein the receiving of the read command, the start read address, and the data chunk indicator includes: receiving the page read command and the start read address; and receiving the data read command and the data chunk indicator.
- 17.** The method of claim 16, wherein the page read command includes a first read command and a second read command; and the data read command includes a third read command and a fourth read command.
- 18.** The method of claim 17, wherein the receiving the page read command and the start read address includes sequentially receiving the first read command, the start read address, and the second read command; and the receiving the read command and the data chunk indicator includes sequentially receiving the third read command, the data chunk indicator, and the fourth read command.
- 19.** The method of claim 15, wherein the outputting one or more of the plurality of first data chunks includes continuously outputting two or more of the plurality of first data chunks indicated by the data chunk indicator.
- 20.** A memory system comprising: a memory controller configured to output a read command, a start read address, and a data chunk indicator, the read command including a page read command

and a data read command, and the start read address including a start row address and a start column address; and a nonvolatile memory device configured to operate based on the read command, the start read address, and the data chunk indicator, the nonvolatile memory device including, a memory cell array configured to output first page data based on the start row address, the first page data including a plurality of first data chunks, and a page buffer circuit configured to continuously output two or more of the plurality of first data chunks based on the data chunk indicator.
